

Practices Card

The operational habits that decide whether a small model becomes a service or stays a demo. One page, in the order you will need them.

01 · EVALUATION

Compose the set, do not sample it

SLICE	SHARE
Representative traffic	50%
Known hard cases	20%
Adversarial	15%
Boundary	10%
Should refuse or escalate	5%

Every production incident becomes a permanent case. Score per slice, because the aggregate hides the failures that matter.

02 · EVALUATION

Cheapest check that works

- **Schema and type checks.** Free, deterministic, catches most pipeline breakage.
- **Exact or normalised match.** Free, needs a known answer.
- **Programmatic properties.** No invented identifiers, totals reconcile, citations resolve.
- **Similarity to a reference.** Cheap and noisy. A signal, never a gate.
- **Model as judge.** Expensive and biased. Last resort, and only once validated.

03 • EVALUATION

Validate the judge or do not use it

- Hand label 50 cases and measure agreement. Below about **80 percent** it cannot be a gate.
- Control position bias by running both orders. Control length bias. Control self preference.
- **Never let a model judge its own family** in a gate.
- Pin the judge and its version. A silently upgraded judge invalidates your history.

04 • EVALUATION

Every number needs an interval

On 100 cases at 80 percent, the 95 percent interval is roughly **plus or minus 9 points**. A move from 80 to 84 is not evidence.

- Interval is about $2 \times \sqrt{p(1-p)/n}$.
- For two models, compare **per case**, not per aggregate. Far more sensitive.
- Fix seed and temperature, or you are measuring sampling noise too.

05 • MIXTURE OF EXPERTS

Memory follows total, speed follows active

SHAPE	MEMORY LIKE	SPEED LIKE
8B dense	8B	8B
30B total, 3B active	30B	~3B
20B total, 4B active	20B	~4B

If you own the memory, it is a very good deal. If memory is your binding constraint, it is the worst of both worlds.

06 • MIXTURE OF EXPERTS

What only shows up in production

- Load time follows total size, so it is slow to start and slow to swap. Keep it resident.
- Expert offloading costs **tail** latency, not average.
- Batching is less predictable when a batch activates many experts at once.
- Benchmark at your real concurrency. Batch size one flatters it.

07 · MEMORY

Weights are the part you budgeted for

`kv ~= 2 x layers x kv_heads x head_dim x context x batch x bytes`

- Cache grows with context **and** concurrency. They are one budget.
- Long context and high concurrency are the same problem in different clothes.
- Reserve **20 percent headroom** for fragmentation, or you will fail to allocate after hours of traffic.

08 · MEMORY

Levers, cheapest first

- Send less context. Usually free and usually improves quality.
- Cap the context limit, then guard long inputs so the answer is never truncated.
- Quantize the cache to 8 bit. Roughly halves it.
- Reduce concurrency. Then quantize weights further. Then offload.
- Use a smaller model, which is the honest answer more often than people admit.

09 · MEMORY

Get the prefix back

- Order the prompt: **stable first**, volatile last.
- Keep the prefix byte stable. **One timestamp at the top discards the whole cache.**
- Pin a session to one worker, or the cache you built is on another machine.

Measured in the lab: 0.15s of prompt processing became 0.04s when the prefix was reused, and stayed at 0.14s when a timestamp broke it.

10 · PRODUCTION

Version prompts like code

- Prompts live in the repository, reviewed, and gated by the eval on every change.
- Log **prompt version, model id and decoding parameters** together. All three define behaviour.
- If you cannot say what produced an output six weeks ago, you have a long running demo.

11 • PRODUCTION

Fail without amplifying

- **Token cap on every call.** An unbounded generation is an unbounded outage.
- Timeouts shorter than caller patience. Retries bounded and jittered.
- **Change something on retry.** An identical call at temperature zero returns the identical failure.
- Circuit break to the deterministic path. Make escalation a budget, not a fallback.

12 • PRODUCTION

Upgrades are migrations

- Frozen golden set, identical harness and conditions.
- Compare **per case**. Find what broke, not whether the mean moved.
- Shadow on live traffic, then canary behind the same gates.
- Keep the old model deployable for a full traffic cycle.
- Prompts are tuned to a model. Unchanged prompt plus new model is a hope, not an experiment.

13 • PRODUCTION

Security, short version

- **Treat every model output as untrusted input** to the next stage. Always.
- Never let model output authorise an action. Tools carry their own authorisation.
- Redact before logging. Scrub before training. Both are irreversible afterwards.
- Pin model versions and digests. Record the licence per model.

14 • THE ONE THAT OUTRANKS THE REST

Cut N before you buy parameters

PER STEP	5 STEPS	10 STEPS	20 STEPS
0.90	59%	35%	12%
0.95	77%	60%	36%
0.99	95%	90%	82%

Reliability comes from architecture, not parameters. Decompose, verify, then buy the cheapest model that clears the bar.